

UNIVERSAL SUPPLIER QUALITY & FAI COMPLIANCE SOP

Document Reference: SOP-SQA-2026-011 | Standard Operating Procedure

1. OBJECTIVE & UNIVERSAL SCOPE

This Standard Operating Procedure defines a universal framework for onboarding new suppliers, evaluating First Article Inspection (FAI) submissions, and enforcing structured recovery protocols when non-conformances occur. Designed as an industry-agnostic standard, this document applies universally across manufacturing, automotive, aerospace, precision engineering, and tech hardware sectors.

2. AUTOMATED & ISO-BASED DUAL SENSOR GRADING SYSTEM

To ensure objective, real-time evaluation of incoming materials and supplier performance, a dual-layer sensor-based monitoring system is integrated into the inspection framework:

- **Layer 1 (Automated Operational Sensors):** Processes operational and dimensional data as components pass through initial inspection. The system automatically categorizes suppliers into three distinct risk tiers:
 - **Approved Supplier:** Granted upon 100% compliance during FAI with zero critical or major defects.
 - **Medium Risk Supplier:** Identified when minor non-critical deviations are detected that can be rectified rapidly without structural impact.
 - **High Risk Supplier:** Triggered immediately upon FAI rejection, major dimensional out-of-tolerance conditions, or material specification failures.
- **Layer 2 (ISO-Based Smart Compliance Sensors):** Performs automated scanning of physical material properties, surface finishes, and geometry directly against ISO standard specifications. This eliminates human inspection error and ensures absolute technical verification against international standards.

3. FAI REJECTION & MANDATORY 30-DAY SCAR RECOVERY PROTOCOL

When a new supplier fails the First Article Inspection (FAI), a mandatory Supplier Corrective Action Request (SCAR) protocol is initiated. The top three root causes identified via Pareto Analysis are subjected to a strict 30-day recovery timeline divided into three 10-day phases.

30-Day Execution Matrix (10 Days Per Root Cause):

- **Phase 1 (Days 01–10):** Corrective action implementation and daily verification for Root Cause 1.
- **Phase 2 (Days 11–20):** Corrective action implementation and daily verification for Root Cause 2.
- **Phase 3 (Days 21–30):** Corrective action implementation, daily verification for Root Cause 3, and final FAI resubmission.

Daily Documentation Requirements:

During each 10-day block, the supplier must submit systematic daily progress logs. The documentation must explicitly detail:

- Specific measures and corrective actions implemented on each day.
- Quality engineering tools, methodologies, and techniques utilized to rectify the root cause and improve the supplier rating.
- Daily measurable outputs and learnings applied toward achieving FAI resubmission readiness.

4. FINAL SIGN-OFF & APPROVAL AUTHORITY

Designated Authority: Head of Operations

Technical Justification: The authority to approve or reject a First Article Inspection report is assigned exclusively to the Head of Operations. Rejecting an FAI requires extensive industrial expertise and deep technical insight to validate non-conformance claims.

Furthermore, following a Toyota-style supplier partnership philosophy, the Head of Operations does not merely reject submissions, but provides detailed guidance, feedback, and actionable corrective measures to enable the supplier to elevate their process capability and successfully pass resubmission.

TQM TOOLS APPLIED	ISO STANDARDS APPLIED
• Pareto Analysis (Root Cause Prioritization) • 8D Problem-Solving Framework • Daily Action & Progress Log Matrix • Supplier Corrective Action Request (SCAR) • Risk Classification Matrix	• ISO 9001:2015 (Quality Management Systems) • AS9102 (First Article Inspection Requirement) • ISO Clause 8.4 (Control of Externally Provided Processes)

Notice: This document is created as part of a self-learning practice initiative and professional portfolio development. No real company or proprietary data was utilized. All protocols are designed to reflect high-level industrial standards.